Operations in an orthonormal basis

Each row is an operation that costs a linear system in a general basis and costs a dot product here. The saving is not incremental — the system disappears entirely, because orthogonality makes the coordinates independent of each other.

6
OPERATIONS

COORDINATES — 2

1 Finding a coordinate

one dot product per coordinate

$$c_i = \mathbf{x} \cdot \mathbf{q}_i$$

In a general basis this means solving $B\mathbf{c} = \mathbf{x}$ — all coordinates at once, none available before the rest. Here each is an independent dot product, so any single coordinate can be computed without the others. This is the entry the rest depend on.

2 Reconstruction

over a full orthonormal basis

$$\mathbf{x} = \sum (\mathbf{x} \cdot \mathbf{q}_i)\, \mathbf{q}_i$$

The expansion written out. Each term is the component of $\mathbf{x}$ along one basis direction, so the sum is a decomposition into perpendicular pieces rather than an arbitrary combination that happens to work.

LENGTHS AND ANGLES — 2

3 Squared length

Parseval's identity

$$\|\mathbf{x}\|^2 = \sum c_i^2$$

Pythagoras in n dimensions. The length of a vector is recoverable from its coordinates alone — which fails in a general basis, where the coordinates say nothing about length without the Gram matrix to weight them.

4 Inner product

coordinates in the same basis

$$\mathbf{x} \cdot \mathbf{y} = \sum c_i d_i$$

The inner product of the vectors is the ordinary dot product of their coordinate vectors. Geometry is preserved exactly under the change of coordinates, which is what makes an orthonormal basis a rigid motion rather than a distortion.

PROJECTION ONTO A SUBSPACE — 2

5 Projection

$\{\mathbf{q}_i\}$ orthonormal basis for W

$$\operatorname{proj}_W \mathbf{x} = \sum_{i=1}^{k} (\mathbf{x} \cdot \mathbf{q}_i)\, \mathbf{q}_i$$

No $(A^{\mathsf{T}}A)^{-1}$ anywhere — compare the general projection formula, which needs a matrix inverse. Orthonormality makes $A^{\mathsf{T}}A = I$, so the inverse is the identity and the formula collapses to a sum of independent terms.

6 Bessel's inequality

equality iff $\mathbf{x} \in W$

$$\sum_{i=1}^{k} c_i^2 \le \|\mathbf{x}\|^2$$

A truncated sum can only lose length. The deficit is exactly the squared distance from $\mathbf{x}$ to W, so the inequality measures how much of the vector the subspace fails to capture — which is the least-squares error before any least-squares problem is posed.

One fact produces all of them: $\mathbf{q}_i \cdot \mathbf{q}_j$ is 1 when $i = j$ and 0 otherwise, so every cross term in every expansion vanishes. That is why an orthonormal basis is worth the Gram–Schmidt cost of constructing one — the expense is paid once and every operation afterwards is cheaper.